

Master thesis:

Photochemical modelling of Ozone and O₂(1a_{delta})

Background:

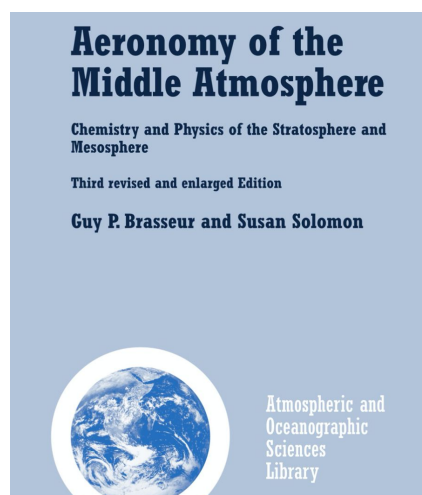
We have satellite measurements of O₂(1a_{delta}) concentration, which is closely linked to the concentration of ozone, thus we can retrieve ozone from this satellite data by using a photochemical kinetic model. The problem is, a lot of measurements that we have from Odin was measured closed to the local sunrise, where the assumption of photochemical equilibrium is hardly valid. Therefore, we need a time-dependent kinetic model, instead of an equilibrium model which was used previously, to adjust our estimates on ozone concentration.

A time-dependent model, or dynamical model, can be simply understood as a model that allows us to see how different chemical species (O₃, OH, O, O₂delta....) concentrations develop over time, given an initial condition. When we have this time-dependent photochemical model working, we can compare our measurements at, say 06:00, to another ozone data from a different satellite which was taken at, say 1500 local time.

Simplified reaction scheme:

In the mesosphere, O₂(1a_{delta}) highly relies on the available O₃. And how much O₃ is affected by the HO_x species which influences the oxygen cycle (O, O₂, O₃).

The following figure is taken from this book, which is more or less our standard text book in this field ;)



O2(1a_delta) - O3

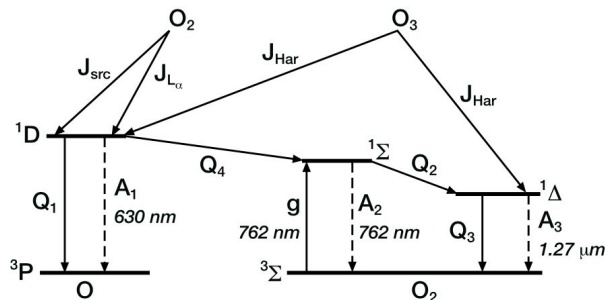


Figure 2.10. Oxygen dayglow production mechanism in the middle atmosphere. The photolysis of ozone (J_{Har}) in the Hartley band ($\lambda < 320$ nm) leads to the formation of the excited oxygen atom $O(^1D)$ and molecule $O_2(^1\Delta_g)$. The photolysis of molecular oxygen in the Schumann-Runge continuum (J_{SRC} at $\lambda < 175$ nm) and at Lyman α ($J_{L\alpha}$ at $\lambda = 121.6$ nm) leads to the formation of 2 electronically excited oxygen atoms $O(^1D)$. De-activation is associated with radiative emissions (A_i) and quenching (Q_i). The emission A_1 produces a red line at 630 nm (see Table 2.1), A_2 produces atmospheric bands at wavelengths near 762 nm, and A_3 (Reaction 2.93) produces infrared bands at 1.27 μm . The quenching processes correspond to the following reactions: 2.90 for Q_1 and Q_4 , 2.92 for Q_2 , and 2.94 for Q_3 . Radiative excitation (g) at 762 nm is represented by reaction 2.91 (from Mlynarczyk *et al.*, 1993).

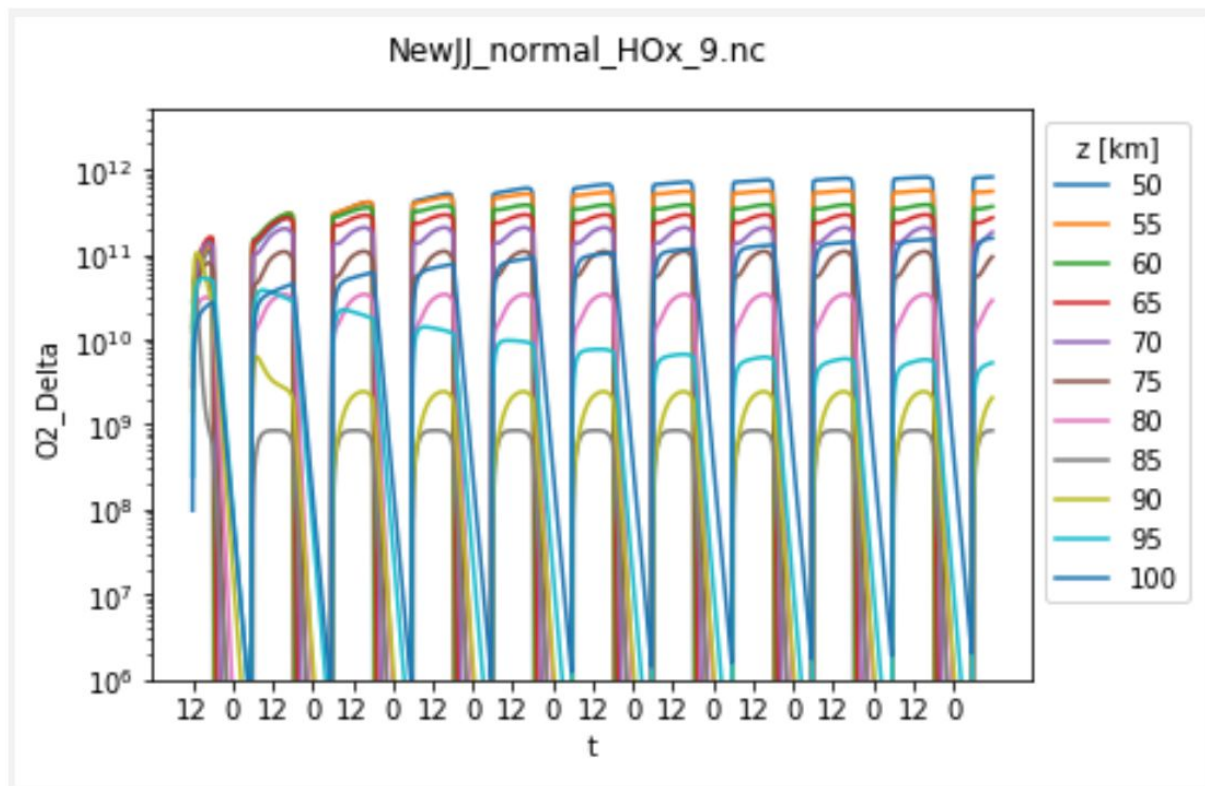
O3 - HOx

TABLE 2. REACTIONS AND RATE COEFFICIENTS

Reaction	Rate coefficient*
$O + O_2 + M \rightarrow O_3 + M$	$k_1 = 1.07 \times 10^{-34} \exp(510/T)$
$O_3 + h\nu \rightarrow O_2 + O$	$J_2 = 9.2 \times 10^{-3} s^{-1}$
$H + O_3 \rightarrow OH + O_2$	$k_3 = 1.0 \times 10^{-10} \exp(-516/T)$
$OH + O \rightarrow H + O_2$	$k_4 = 4.2 \times 10^{-11}$
$H + O_2 + M \rightarrow HO_2 + M$	$k_5 = 2.08 \times 10^{-32} \exp(290/T) \gg$
$OH + O_3 \rightarrow HO_2 + O_2$	$k_6 = 1.5 \times 10^{-12} \exp(-1000/T)$
$HO_2 + O \rightarrow OH + O_2$	$k_7 = 3.5 \times 10^{-11}$

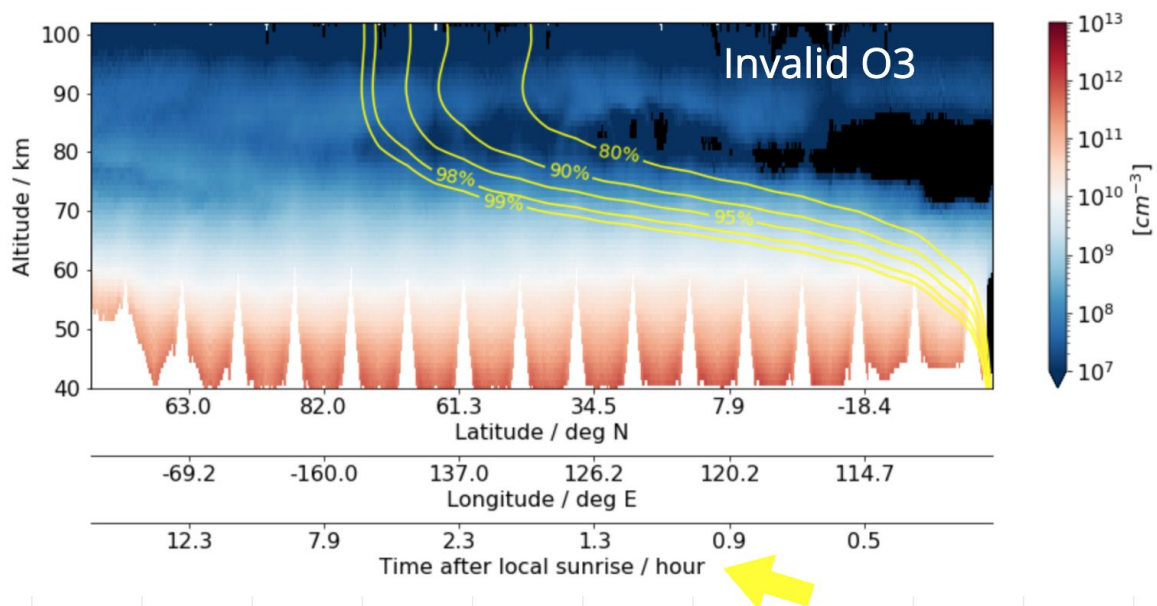
from Frederick 1979

Once the time dynamical model is probably built, it should have an output something like this:



The horizontal axis t has the unit of hour of a day. When the sun rises, the concentration of $O2_Delta$ quickly rises. When the sun sets, the speed of decay depends on altitude. The model runs at an initial value, and then slowly settles down in a repetitive diurnal cycle, so the initial value we put shall not matter so much in the last model day, which is what we want from this model.

If the model is running well, we can make use of the “invalid O3” part of the data, which is indicated in the upper right corner of the following plot.



(The black color in the plot represents negative ozone concentration, which is unphysical! It is the result of using an equilibrium assumption to retrieve ozone from $O2(1a_delta)$)